

CIRROSE NA UTI - 2



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Doutor em Ciências Médicas - HCFMUSP



1 HEMODINÂMICA – PARTE 2

2 RESPIRATÓRIO



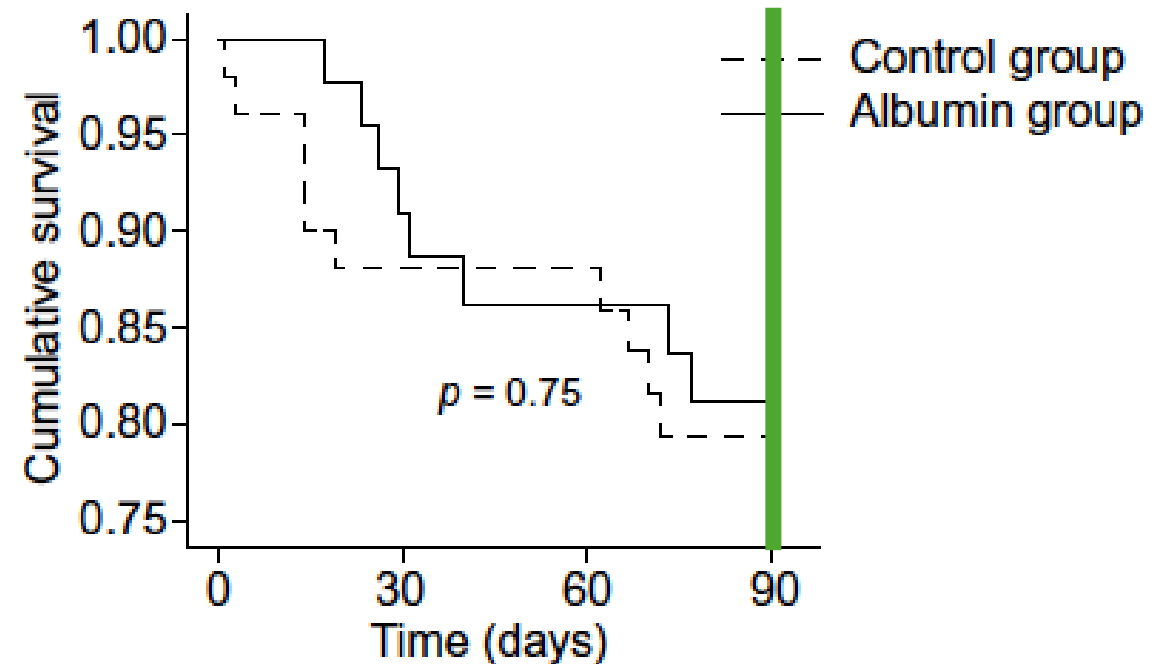
Albumina na Sepses + Cirrose ?

Albumin for bacterial infections other than spontaneous bacterial peritonitis in cirrhosis. A randomized, controlled study[☆]

Mónica Guevara^{1,2,3,4,†}, Carlos Terra^{1,2,3,4,†}, André Nazar^{1,2,3,4}, Elsa Solà^{1,2,3,4}, Javier Fernández^{1,2,3}, Marco Pavesi^{1,2,3}, Vicente Arroyo^{1,2,3}, Pere Ginès^{1,2,3,4,*}

J Hepatol. 2012 Oct;57(4):759-65

Em 2012 ...
N100
Unicêntrico



Qual o melhor fluido na cirrose ?

Hepatology International (2021) 15:983–994
<https://doi.org/10.1007/s12072-021-10164-z>

ORIGINAL ARTICLE



Comparison of 5% human albumin and normal saline for fluid resuscitation in sepsis induced hypotension among patients with cirrhosis (FRISC study): a randomized controlled trial

Cyriac Abby Philips¹ · Rakhi Maiwall¹ · Manoj Kumar Sharma¹ · Ankur Jindal¹ · Ashok Kumar Choudhury¹ · Guresh Kumar² · Ankit Bhardwaj² · Lalita Gouri Mitra³ · Prashant Mohan Agarwal³ · Shiv Kumar Sarin^{1,2} 

ALBUMINA **5%** (250 BOLUS + 50ML/H – 3H)
X
SF 30ML/KG + 100ML/H – 3H

Table 2 Mean arterial pressure response in the two study groups

Mean arterial pressure $\geq$ 65 mm Hg	5% human albumin <i>n</i> (%)	Normal saline <i>n</i> (%)	<i>p</i> value	OR (95% CI)
1 h	39 (25.3%)	23 (14.9%)	0.03	1.9 (1.08–3.42)
2 h	27 (17.5%)	15 (9.7%)	0.06	1.9 (1.00–3.87)
3 h	18 (11.7%)	5 (3.2%)	0.008	3.9 (1.42–10.9)

> RESOLUÇÃO DE HIPOTENSÃO

+ SUSTENTADO ?

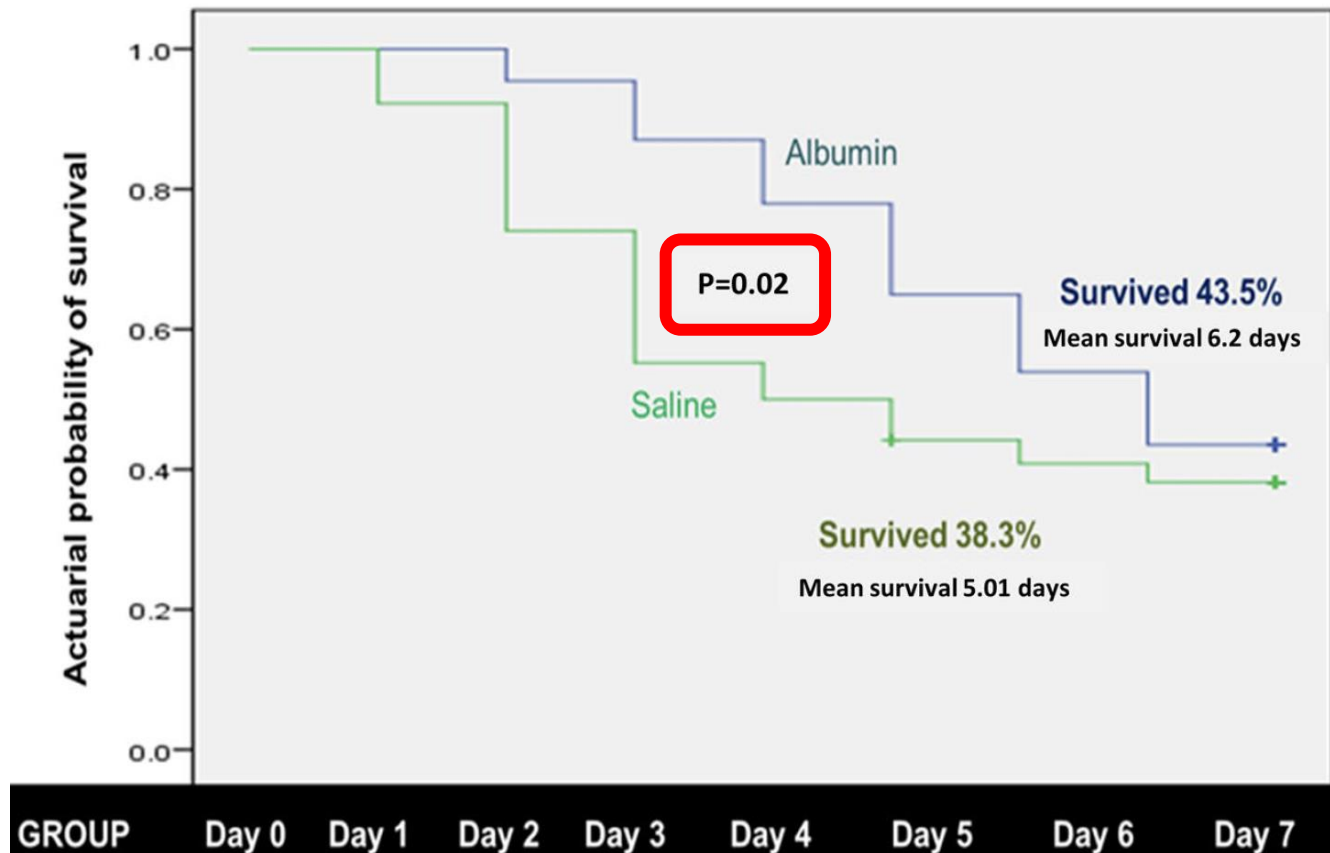


**DESFECHO SECUNDÁRIO
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ABERTO
N 308**

Comparison of 5% human albumin and normal saline for fluid resuscitation in sepsis induced hypotension among patients with cirrhosis (FRISC study): a randomized controlled trial

Cyriac Abby Philips¹ · Rakhi Maiwall¹ · Manoj Kumar Sharma¹ · Ankur Jindal¹ · Ashok Kumar Choudhury¹ · Guresh Kumar² · Ankit Bhardwaj² · Lalita Gouri Mitra³ · Prashant Mohan Agarwal³ · Shiv Kumar Sarin^{1,2}

SURVIVAL FUNCTIONS



ALBUMINA 20% (0.5-1G/KG) - 3H X PL 30ML/KG - 3H

A randomized-controlled trial comparing 20% albumin to plasmalyte in patients with cirrhosis and sepsis-induced hypotension [ALPS trial]

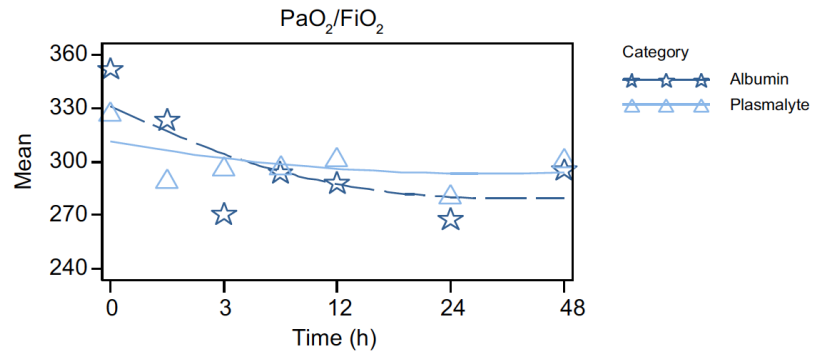
Table 2. Changes in systemic hemodynamics, parameters of microcirculation, biochemical parameters, severity scores and fluid indices between the 2 groups.

Variable	20% albumin (n = 50)	Plasmalyte (n = 50)	*p value	#p value
Parameters of macro and microcirculation				
Mean arterial pressure (mmHg)	51.40 ± 6.52	49.88 ± 4.45	0.18	0.001
30 mins	60.20±5.11	58.28±4.55	0.05	
45 mins	63.98±4.47	60.72±3.87	<0.001	
1 hour	65.28±4.10	62.58±4.21	0.002	
2 hours	66.58±3.34	66.14±5.60	0.63	
3 hours	68.04±2.41	65.68±4.83	0.003	
6 hours	69.78±2.92	70.04±3.35	0.68	
12 hours	69.08±11.20	71.06±4.49	0.25	
24 hours	72.87±12.78	75.71±5.28	0.26	

Arterial lactate (mmol/L)

0 hour		6.16 ± 3.19	6.38±4.77	0.79	<0.001
1 hour		4.28±2.54	5.56±3.95	0.06	
3 hours	> QUEDA LACTATO	2.90±1.60	4.90±3.72	<0.001	
6 hours		2.47±1.99	4.20±3.33	0.002	
12 hours		2.34±2.07	3.84±3.46	0.01	
24 hours		2.04±1.53	3.23±3.36	0.03	
48 hours		1.85±1.68	3.04±3.33	0.03	

PROBLEMAS ...



We had to discontinue albumin in a total of 11 (22%) patients. Six (55%) patients developed adverse effects at 3 hours while 5 (45%) patients did at 6 hours. We noted a concomitant increase in the partial pressure of carbon dioxide levels in albumin group

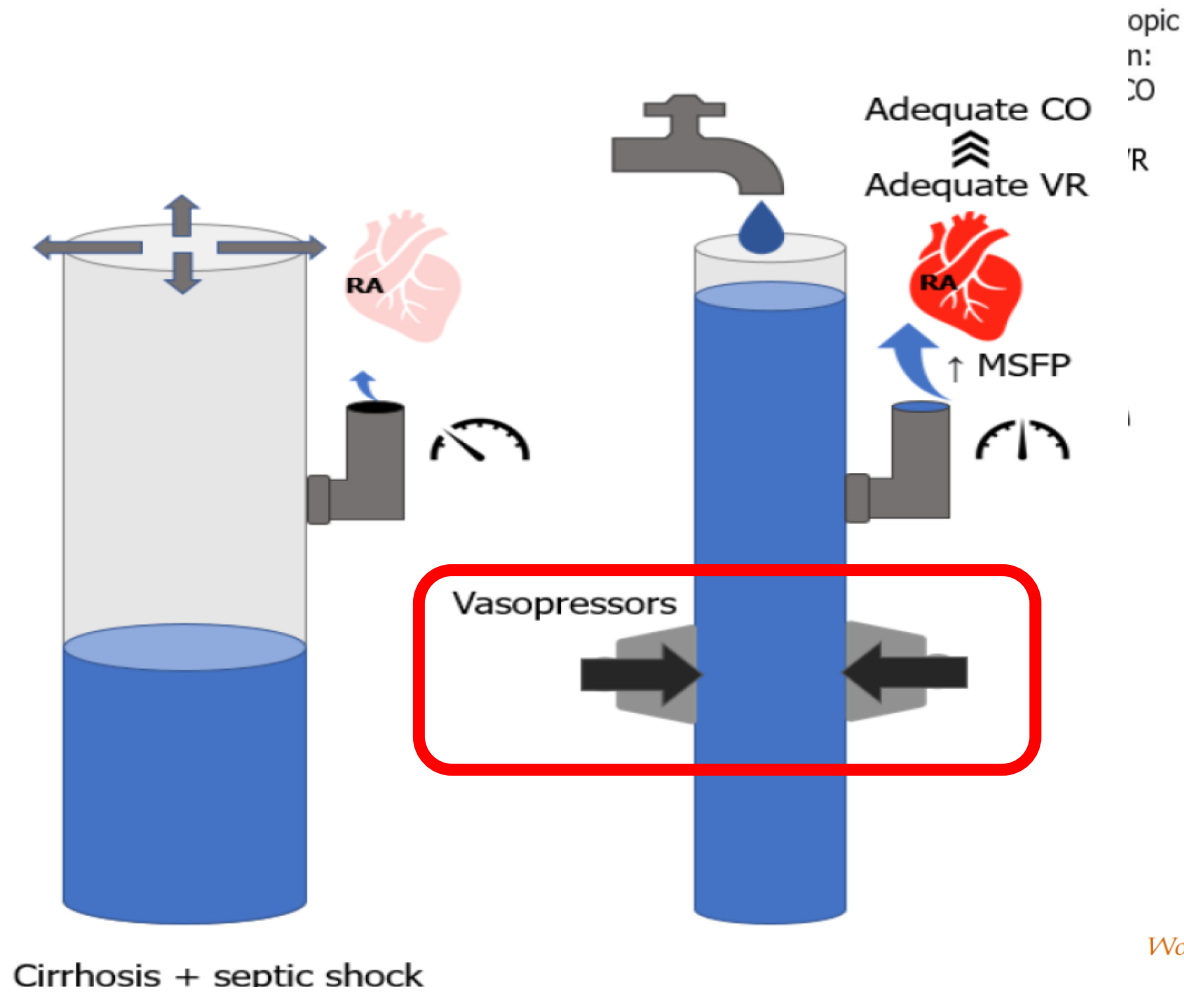
ratio of partial pressure of oxygen on inspired oxygen ($p < 0.001$),

(Table 3). The main side effects of albumin were pulmonary edema in 6 (12%) patients, 1 patient had an allergic reaction with rashes and bronchospasm, 6 (12%) patients had bronchospasm and wheeze, and rebound hypertension in 2 (4%) patients. At the same time, the plasmalyte was safer with few adverse effects.

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VOLUME X VASOPRESSOR

C Restrictive fluids and early vasopressors



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CO
R

World J Hepatol 2023 April 27; 15(4): 497-514

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Vasoplegia x Cirrose

Vasopressin deficiency and vasodilatory state in end-stage liver disease

Gebhard Wagener, MD^{1, #}, Galina Kovalevskaya, PhD², Moury Minhaz, BS¹, Fallon Mattis, MD¹, Jean C. Emond, MD³, and Donald W. Landry, MD, PhD⁴

Correlation of SVRI and preOP MELD

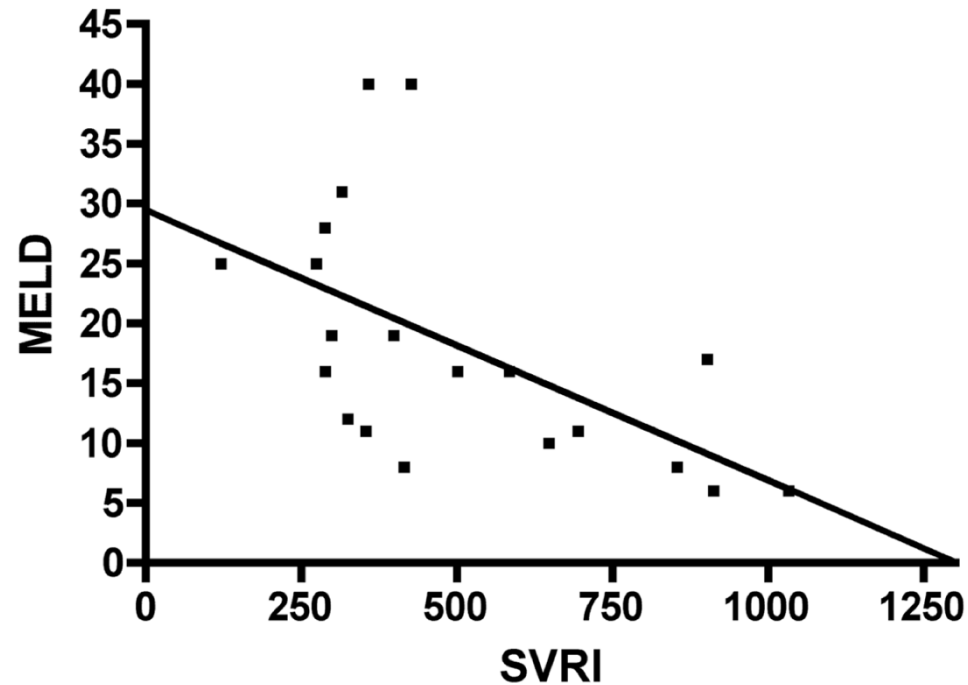


Figure 2.
Correlation of baseline systemic vascular resistance index (SVRI) with preoperative (preOP) Model of Endstage Liver Disease (MELD) score; Pearson $r = -0.567$, $p < 0.01$



Vasopressin deficiency and vasodilatory state in end-stage liver disease

Gebhard Wagener, MD^{1,#}, Galina Kovalevskaya, PhD², Moury Minhaz, BS¹, Fallon Mattis, MD¹, Jean C. Emond, MD³, and Donald W. Landry, MD, PhD⁴

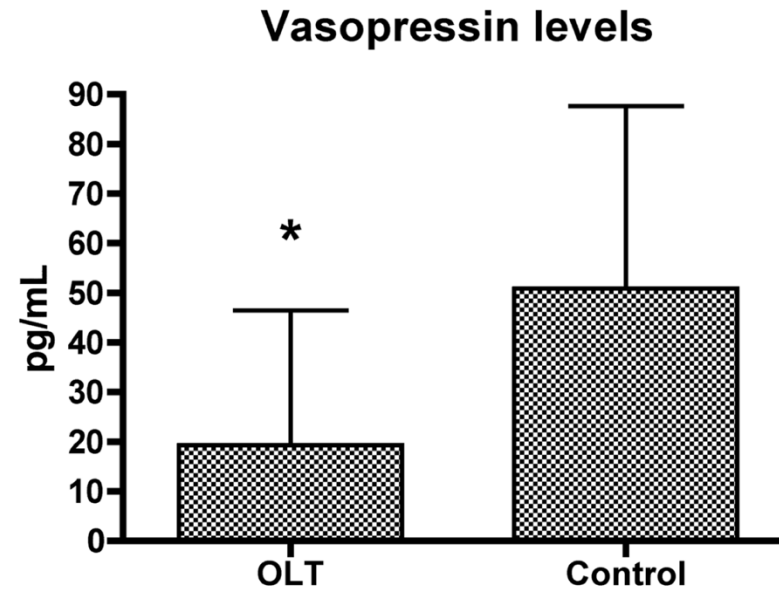


Figure 1. Baseline vasopressin levels in patients with liver disease undergoing liver transplantations (OLT, n=28) and control patients with normal liver function undergoing hepatectomies or Whipple procedures (n=7), (mean $\pm$ SD, * - $p < 0.05$). Four OLT and 4 control patients had baseline vasopressin levels above the detection limit of 80 pg/mL and their levels were defined as 80 pg/mL.



Qual Vasopressor ?

Vasopressin in sepsis and septic shock

T. G. KAMPMEIER, S. REHBERG, M. WESTPHAL, M. LANGE

(*Minerva Anestesiol* 2010;76:844-50)

TABLE I.—*Pharmacological characteristics of arginine vasopressin and terlipressin.*

	Arginine vasopressin	Terlipressin
Elimination half-life	6 min	50 min
Effective half-life	5-20 min	4-6 h
V1/V2-ratio	1.0	2.2
Elimination	Rapid metabolism by L/c-aminopeptidase	Slowly cleavage by endopeptidases

urine and plasma osmolality.¹⁹ TP is a synthetic vasopressin analog (Figure 2). It is cleaved by



Primary outcome

It was to maintain a MAP of >65 mmHg

Cirrhotic suspected of septic shock



LIVER FAILURE, CIRRHOSIS AND ITS COMPLICATIONS

A randomized trial comparing terlipressin and noradrenaline in patients with cirrhosis and septic shock

Ashok Choudhury, Chandan K. Kedarietty, Chitranshu Vashishtha, Deepak Saini, Sachin Kumar, Rakhi Maiwall, Manoj K. Sharma, Ajeet S. Bhadoria, Guresh Kumar, Yogendra K. Joshi, Shiv K. Sarin✉

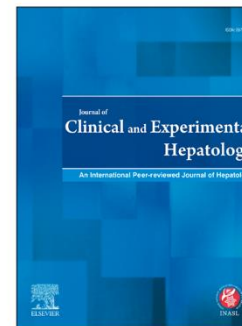
First published: 16 September 2016 | <https://doi.org/10.1111/liv.13252> | Citations: 53

PARAMETERS	Terlipressin Group (n=42)	Noradrenaline group (n=42)	P value
At 48hrs: Primary Outcome			
MAP>65 mm Hg	39/42(92.9%)	29/42(69.1%)	0.005
Reversal of shock	14/42(33.3%)	5/42(11.9%)	0.02

Adverse Effects	Terlipressin Group (n=42)	Noradrenaline group (n=42)	P value
Incidence of adverse effects	17(40.5%)	9(21.4%)	0.06

“Comparative study of Terlipressin and Noradrenaline as vasopressors in patients with ACLF and septic shock: A Randomized Controlled Trial”

Tarana Gupta, MBBS, MD, DM Hepatology, Professor, Anjali Saini, MBBS, MD, Resident, Vaibhav Gaur, MBBS, MD, Resident, Ashank Goel, MBBS, MD, Resident



“Comparative study of Terlipressin and Noradrenaline as vasopressors in patients with ACLF and septic shock: A Randomized Controlled Trial”

METHODS



70 Patients
ACLF with septic shock

Parallel group,
Open label,
Randomized (1:1)
single centre trial

INTERVENTION



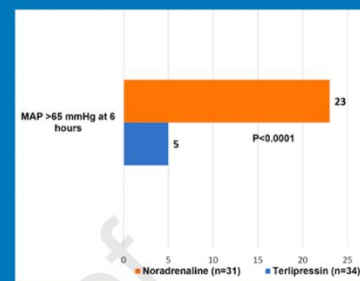
Terlipressin
(n=34)

Noradrenaline
(n=31)

PRIMARY OUTCOME

**Achievement of
Target MAP >65
mmHg at 6 hours**

RESULTS
PRIMARY OUTCOME



Secondary outcomes

- 3-and 7-day mortality was significantly higher in terlipressin as compared to noradrenaline group
- No difference in mean cumulative dose of drug, duration of hospital stay
- New onset events of UGIB, AKI, HE were comparable

Nora foi melhor para atingir PAM

Menor mortalidade ????



Qual Alvo de PAM ???

CIRROSE + CHOQUE SÉPTICO + IRA

ALVO DA PAM

- 60-65 mmHg
- 80-85 mmHg

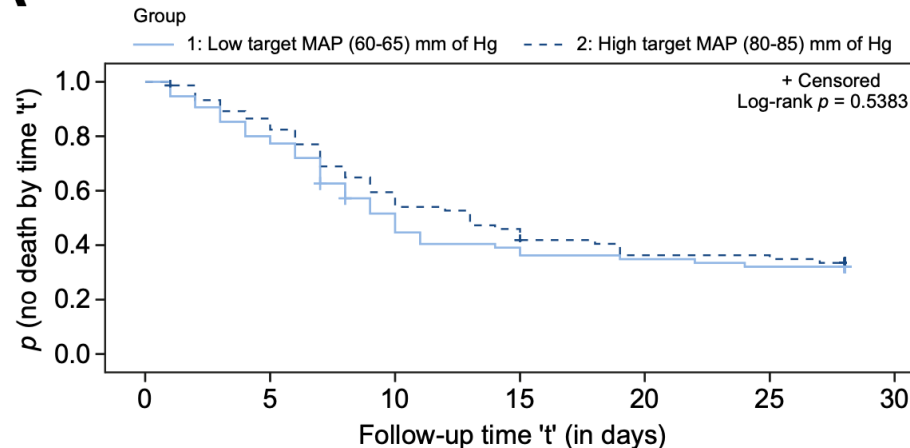
A randomised-controlled trial (TARGET-C) of high vs. low target mean arterial pressure in patients with cirrhosis and septic shock

Rakhi Maiwall¹, Samba Siva Rao Pasupuleti^{2,7}, Ashini Kumar Hidam³, Anupam Kumar³, Harsh Vardhan Tevethia¹, Rajan Vijayaraghavan¹, Arpita Majumdar¹, Adarsh Prasher³, Sherin Thomas⁴, Rajendra Prasad Mathur², Guresh Kumar⁸, Shiv Kumar Sarin^{1,4}

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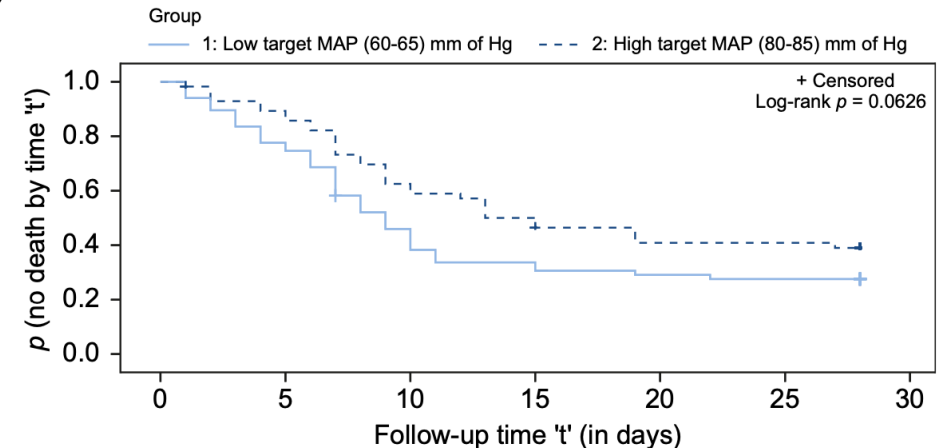
A



N° at risk							
— 1	75	60	37	28	25	23	0
- - - 2	75	64	44	34	26	26	0

B

Per Protocol



N° at risk							
— 1	67	52	30	22	19	18	0
- - - 2	57	50	35	28	22	22	0

A randomised-controlled trial (TARGET-C) of high vs. low target mean arterial pressure in patients with cirrhosis and septic shock

Rakhi Maiwall¹, Samba Siva Rao Pasupuleti^{2,7}, Ashini Kumar Hidam³, Anupam Kumar³, Harsh Vardhan Tevethia¹, Rajan Vijayaraghavan¹, Arpita Majumdar¹, Adarsh Prasher³, Sherin Thomas⁴, Rajendra Prasad Mathur⁵, Guresh Kumar⁶, Shiv Kumar Sarin^{1,*}

Journal of Hepatology 2023. vol. 79 | 349–361



Table 2. Differences between the two randomisation groups for secondary outcome measures for intention-to-treat and per-protocol analysis.

Secondary outcome measures	Intention-to-treat analysis			Per-protocol analysis		
	Low MAP (n = 75)	High MAP (n = 75)	<i>p</i> value	Low MAP (n = 67)	High MAP (n = 57)	<i>p</i> value
AKI recovery	23 (31)	34 (45%)	0.064	21 (31)	30 (53)	0.016
Renal replacement therapy	40 (53)	43 (57%)	0.62	34 (51)	28 (49)	0.86
Intradialytic hypotension*	19/40 (48)	5/43 (12)	<0.001	18/34 (53)	1/28 (4)	<0.001

Corticoide ? Síndrome Hepatoadrenal ?

Table 1 Prevalence of adrenal insufficiency in critically ill patients with liver cirrhosis

Ref.	No. of patients (type of cirrhosis)	Diagnosis and definition of AI	Prevalence of AI
Harry <i>et al</i> ^[14]	20 (ALF/CLD)	SD-SST: Peak cortisol < 500 nmol/L ¹	69%
Marik <i>et al</i> ^[12]	340 (ALF: 24) (CLD: 146) (recent LT: 119) (remote LT: 51)	LD-SST: Peak cortisol < 552 nmol/L or random cortisol level < 414 nmol/L in non-stressed patients or random cortisol level < 552 nmol/L in stressed patients	72% 33% 66% 92% 61%
Tsai <i>et al</i> ^[8]	101 (cirrhosis+ severe sepsis)	SD-SST: Baseline cortisol < 414 nmol/L or delta cortisol < 250 nmol/L if baseline cortisol between 414 and 938 nmol/L	51%
Fernandez <i>et al</i> ^[13]	25 (cirrhosis + septic shock)	SD-SST: Baseline cortisol < 414 nmol/L or delta cortisol < 250 nmol/L if baseline cortisol between 414 and 966 nmol/L	68%
Thierry <i>et al</i> ^[64]	14 (cirrhosis + septic shock)	SD-SST: Baseline cortisol < 414 nmol/L; delta cortisol < 250 nmol/L	77%
du Cheyron <i>et al</i> ^[65]	50 (critically ill cirrhosis)	SD-SST: Baseline cortisol < 414 nmol/L; delta cortisol < 250 nmol/L if baseline cortisol between 414 and 938 nmol/L	82%
Vasu <i>et al</i> ^[86]	24 (critically ill cirrhotics)	SD-SST: Definition of AI was not reported	62%
Arabi <i>et al</i> ^[29]	75 (cirrhosis + septic shock)	SD-SST: Delta cortisol < 250 nmol/L	76%
Mohamed <i>et al</i> ^[85]	15 (cirrhosis+septic shock)	SD-SST: Definition of AI was not reported	87%
Thevenot <i>et al</i> ^[74]	30 (cirrhosis + sepsis)	SD-SST: Peak serum total cortisol < 510 nmol/L	10%
Acevedo <i>et al</i> ^[89]	166 (decompensated cirrhosis)	SD-SST: Delta cortisol < 250 nmol/L	26%
Graupera <i>et al</i> ^[20]	37 (severe acute bleeding)	SD-SST: Baseline cortisol < 414 nmol/L and/or delta cortisol < 250 nmol/L	38%
Triantos <i>et al</i> ^[16]	20 (cirrhosis with variceal bleeding)	SD-SST: Baseline cortisol < 276 nmol/L or delta cortisol < 250 nmol/L LD-SST: Peak serum cortisol < 690 nmol/L or a delta cortisol < 250 nmol/L	30% 60%
El Damarawy <i>et al</i> ^[66]	45 (cirrhosis with septic shock or HRS, cirrhosis without septic shock or HRS)	SD-SST: Baseline cortisol < 414 nmol/L or delta cortisol < 250 nmol/L in patients with baseline cortisol < 966 nmol/L	73%

¹To convert serum total cortisol concentrations from nanomoles per liter to micrograms per deciliter divide by 27.59^[79]. ALF: Acute liver failure; CLD: Chronic liver disease; HRS: Hepatorenal syndrome; LT: Liver transplant; AI: Adrenal insufficiency; SD-SST: Standard dose short synacthen test; LD-SST: Low dose short synacthen test.



ANTIBIOTICOTERAPIA



O que cobrir ? Onde pesquisar ?

Table 1. (continued)

Variables	20% Albumin (n = 50)	Plasmalyte (n = 50)	p value
Type of infection			
Gram-negative	25 (89.3)	25 (83.3)	0.10
Gram-positive	2 (7.1)	0 (0)	
Fungal	1 (3.6)	5 (16.7)	
Culture-positive infections	28 (56)	30 (60)	
Site of culture-positive infection (n = 58)			
Urine	6 (21.4)	5 (16.7)	
Broncho-alveolar lavage	13 (46.4)	20 (66.7)	
Ascitic fluid	5 (17.9)	3 (10.0)	
Blood	3 (10.7)	0 (0)	
Pleural fluid	0 (0)	1 (3.3)	
Skin and soft tissue	4 (14.3)	1 (3.3)	
Infection at more than 1 site	7 (25)	9 (30)	

PERITONITE BACTERIANA ESPONTÂNEA

- ➔ Infecção da ascite sem causa específica
- ➔ 10% dos Hospitalizados (pesquisar !)
- ➔ > 250 PMN
 - PBE com cultura +
 - PBE com Cultura –
 - Bacterascite ???
 - 5-7 Dias de ATB



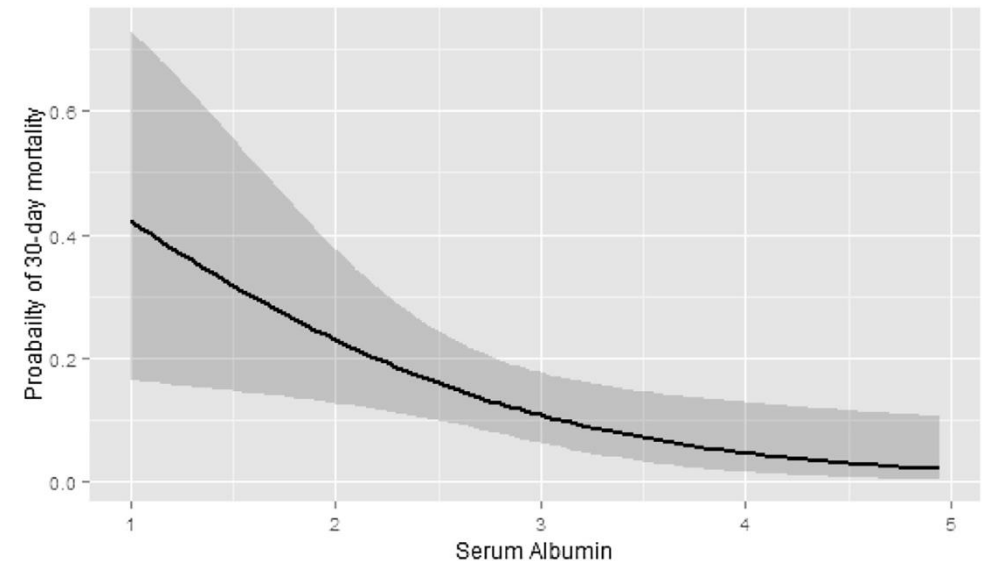
Association between hypoalbuminemia and mortality among subjects treated with ertapenem versus other carbapenems: prospective cohort study

O. Zusman¹, L. Farbman¹, Z. Tredler¹, V. Daitch¹, A. Lador¹, L. Leibovici^{1,2} and M. Paul^{2,3}

Evitar Alta Ligação Protéica ?

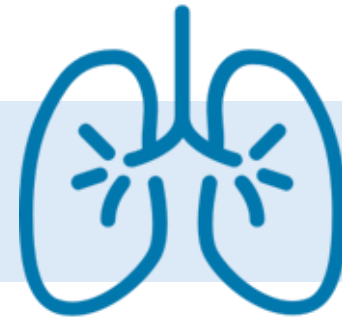
dL). Hypoalbuminemia had a significantly stronger association with mortality among subjects treated with ertapenem compared to I/M, and the association was highly significant only in subjects treated with ertapenem. We observed higher crude

FIG. 1. Probability of 30-day mortality by albumin level in ertapenem-treated subjects. Probability indicated by solid line, confidence interval by shading.



highly protein-bound antibiotics such as ceftriaxone, aztreonam, or carbapenems, increases the unbound fraction and increases its clearance

RESPIRATÓRIO



HIDROTÓRAX HEPÁTICO

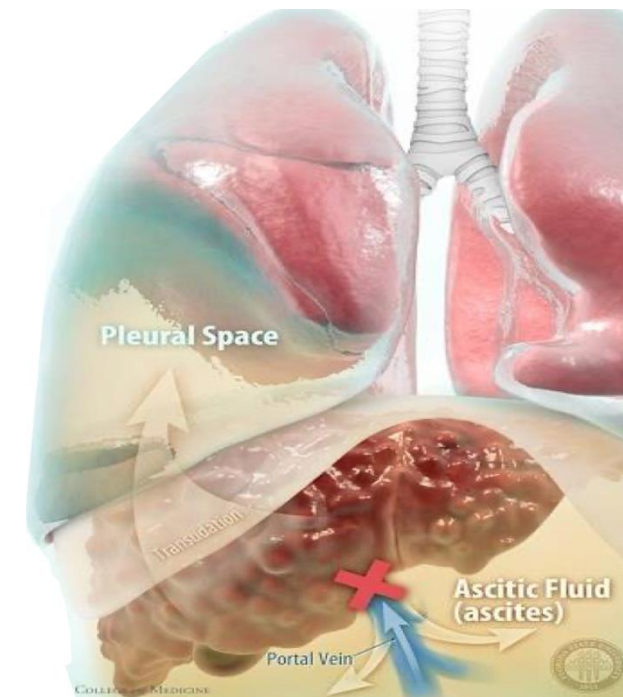
→ Terapia Medicamentosa

→ Toracocentese x Drenos

→ TIPS

→ Cirurgia ?

→ Tx



Effects of PEEP on Liver Arterial and Venous Blood Flows

NICOLA BRIENZA, JEAN-PIERRE REVELLY, TAKAO AYUSE, JAMES L. ROBOTHAM

Pulmonary Anesthesia Laboratory, Department of Anesthesia and Critical Care Medicine, Johns Hopkins Medical Institutions, Baltimore, Maryland

AMERICAN JOURNAL OF RESPIRATORY AND CRITICAL CARE MEDICINE VOL 152 1995

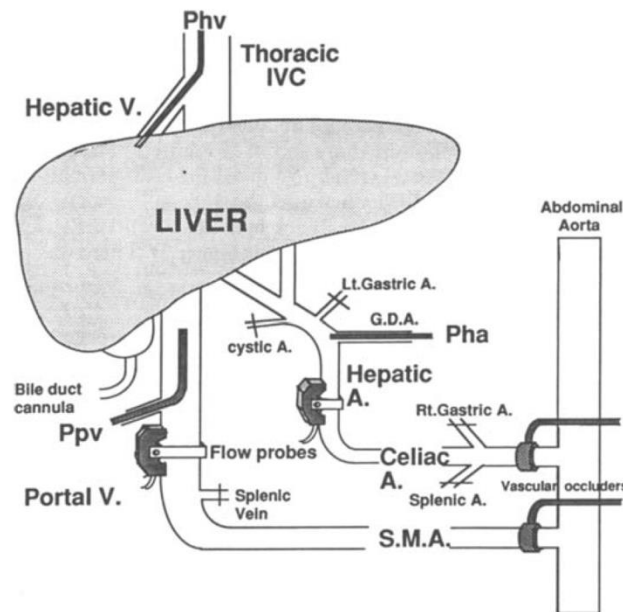


TABLE 1
SYSTEMIC AND HEPATIC HEMODYNAMIC PARAMETERS*

	ZEEP	PEEP 5	PEEP 10	PEEP 15	ZEEP Control
CO	110 ± 6	95 ± 6 [†]	81 ± 7 ^{†,‡}	63 ± 5 ^{†,‡,§}	110 ± 8
HR	112 ± 4	122 ± 5	133 ± 6 [†]	154 ± 8 ^{†,‡,§}	121 ± 6.5
Pao	87 ± 3	85 ± 3	72 ± 4 ^{†,‡}	63 ± 5 ^{†,‡}	81 ± 3.5
SVRI	0.77 ± 0.05	0.87 ± 0.09	0.84 ± 0.08	0.86 ± 0.05	0.72 ± 0.05
Ppa	18 ± 1.2	20.7 ± 1.6 [†]	22.9 ± 1.4 ^{†,‡}	25.4 ± 1 ^{†,‡,§}	18.7 ± 1.1
Pra	4.0 ± 0.2	5.2 ± 0.2 [†]	6.8 ± 0.4 ^{†,‡}	8.6 ± 0.5 ^{†,‡,§}	4.3 ± 0.2
Ppv	6.7 ± 0.3	8 ± 0.3 [†]	9.4 ± 0.3 ^{†,‡}	11.5 ± 0.3 ^{†,‡,§}	7.2 ± 0.5
Pabd	3.6 ± 0.2	4.4 ± 0.3 [†]	5 ± 0.3 ^{†,‡}	6.7 ± 0.4 ^{†,‡,§}	4 ± 0.2
Qpv	21.2 ± 2.6	18.2 ± 2	14.9 ± 2.4 ^{†,‡}	10.8 ± 1.3 ^{†,‡,§}	22 ± 3.4
Qha	9.7 ± 1.7	8.3 ± 1.7	7.4 ± 1.6	5.8 ± 1 ^{†,‡}	12.2 ± 2.3
Qf _{liv}	27.7 ± 2	27.9 ± 2.1	27.1 ± 1.9	26.5 ± 2.2	30.8 ± 1.7

* Mean ± SEM (n = 8). CO, QPV, and Qha = ml/min/kg; Qf_{liv} = %; HR = beats/min; Pao, Ppa, Pra, Ppv, Pabd = mm Hg; SVRI = mm Hg/ml/kg.

[†] p < 0.05 versus value at ZEEP.

[‡] p < 0.05 versus PEEP 5.

[§] p < 0.05 versus PEEP 10.

Effects of PEEP on splanchnic hemodynamics and blood volume

Y. FUJITA

Department of Anesthesiology, Kawasaki Medical School, Kurashiki, Japan

Table 2

Changes in blood gas analysis and hepatic oxygen supply during PEEP₀, PEEP₅ and PEEP₁₀

	PEEP ₀	PEEP ₅	PEEP ₁₀
paH	7.393 ± 0.029	7.383 ± 0.029	7.380 ± 0.028
Paco ₂ kPa	4.8 ± 0.5	4.9 ± 0.4	4.8 ± 0.4
Pao ₂ kPa	19.3 ± 3.9	22.1 ± 2.3	23.6 ± 1.2*
Phvo ₂ kPa	5.5 ± 1.3	5.9 ± 1.5	5.0 ± 0.9†
Shvo ₂ %	62.3 ± 11.9	62.6 ± 17.4	55.1 ± 14.3*†
HDo ₂ ml · min ⁻¹ · 100 g ⁻¹	14.6 ± 2.2	12.7 ± 1.6*	9.5 ± 2.0*†

Data are presented as means ± s.d. Abbreviations are as follows: paH = arterial blood pH, Paco₂ = arterial blood carbon dioxide tension, Pao₂ = arterial blood oxygen tension, Phvo₂ = hepatic venous oxygen tension, Shvo₂ = hepatic venous hemoglobin oxygen satu-

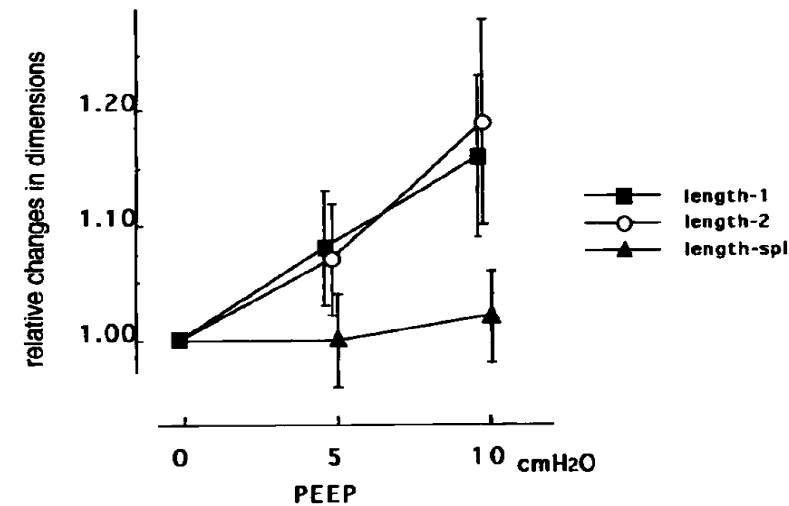


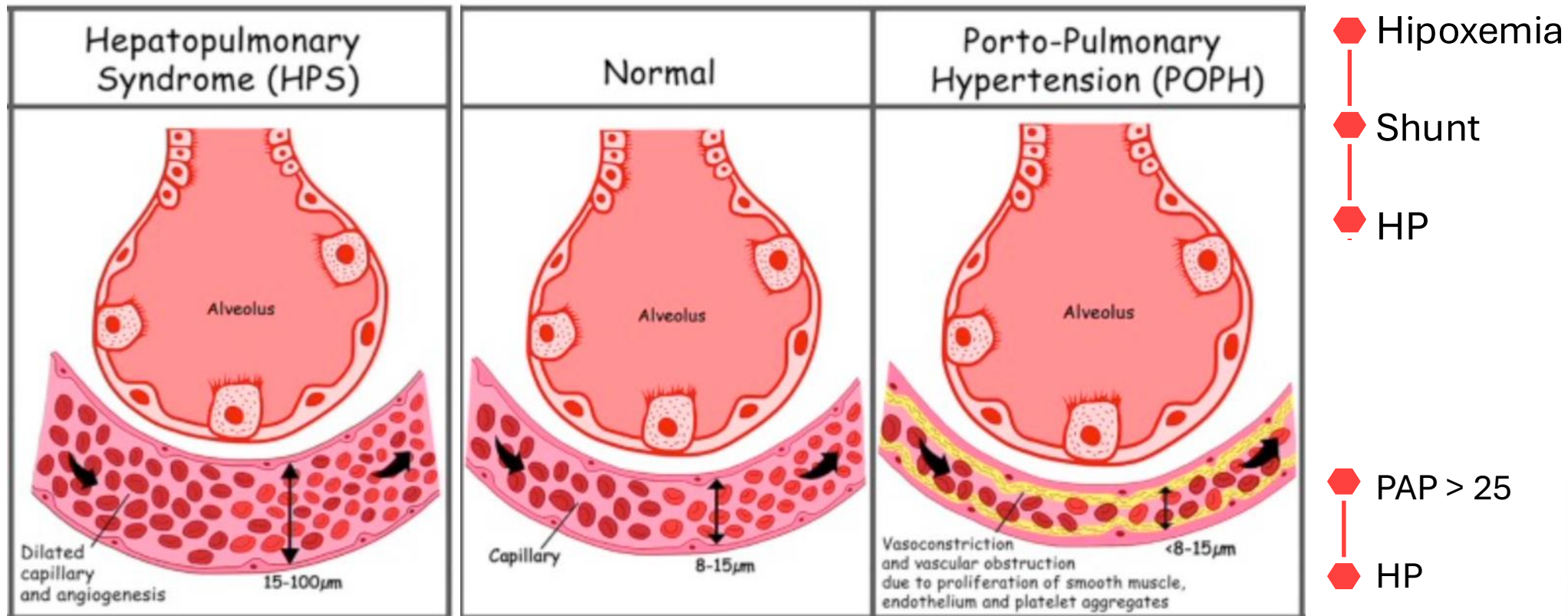
Fig. 2. Changes in hepatic and splanchnic dimensions during 0, 5 and 10 cmH₂O (0, 0.5 and 1.0 kPa) of PEEP. Hepatic dimensions increased with each increase in PEEP from 0 to 10 cmH₂O (0, 0.5 and 1.0 kPa), while splenic dimensions remained unchanged. The values were normalized using the values at PEEP₀ as 1.0. Abbreviations: length-1 = anteroposterior dimension of the middle hepatic lobe, length-2 = transverse dimension of the left hepatic lobe, and length-spl = short axis dimension of the spleen.

Queda na Oxigenação local

Aumento do tamanho (Congestão)



HEPATOPULMONAR X PORTOPULMONAR



HIPOXEMIA NA HEPATOPULMONAR

Methylene Blue Improves the Hepatopulmonary Syndrome

Peter Schenk, MD; Christian Madl, MD; Shahrzad Rezaie-Majd, MD; Stephan Lehr; and Christian Müller, MD



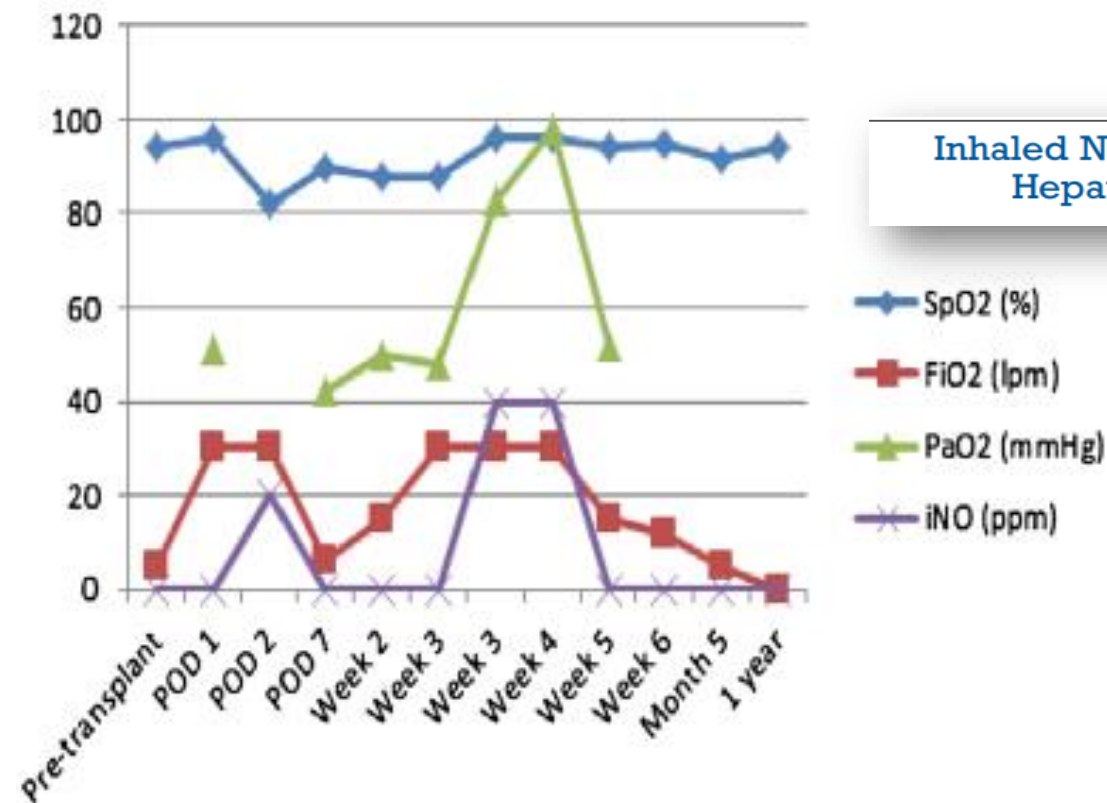
Table. Gas Exchange and Hemodynamic Responses to Short-Term Infusion of Methylene Blue*

Measurement	Baseline	Time after Methylene Blue Infusion							P Value†
		30 Minutes	1 Hour	2 Hours	3 Hours	4 Hours	5 Hours	10 Hours‡	
Gas exchange									
PaO ₂ , mm Hg	58 ± 2.5	69 ± 3.2	71 ± 6	73 ± 10.4	71 ± 11.4	73 ± 12.4	74 ± 11.5	67 ± 11.9	0.006
PAO ₂ – PaO ₂ , mm Hg	49 ± 3.3	38 ± 4.4	34 ± 5.2	33 ± 8.4	33 ± 9	31 ± 11.2	30 ± 10.4	37 ± 9.3§	0.003
Paco ₂ , mm Hg	34 ± 2.7	33 ± 3.7	35 ± 3.4	35 ± 4.3	36 ± 3.9	37 ± 5.9	37 ± 4.5	37 ± 6.7	0.007
Shunt fraction, %	41 ± 3.1	28 ± 4.5	25 ± 3.4	25 ± 4.2	26 ± 4.1	26 ± 6.3	25 ± 4.5	31 ± 14	<0.001



Inhaled nitric oxide improves the hepatopulmonary syndrome: a physiologic analysis

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Inhaled Nitric Oxide as a Potential Rescue Therapy for Persistent Hepatopulmonary Syndrome After Liver Transplantation

consecutive sample of 26 pre-transplant patients. Inhaled nitric oxide significantly improved partial pressure of oxygen (12.4 mm Hg; $p=0.001$) without deleterious effects on cardiac output.



TO BE CONTINUED...

